

Project LOOP — AI Customer-Feedback Intelligence Platform

An Enterprise-Grade Multi-Tenant Customer Feedback Aggregator, Trend Spike Detector, and Grounded RAG Assistant

Developers (Group): M. Ranjith Kumar, M. Renuka Bindu

Internship Track: Web Development

GitHub Repository: <https://github.com/23A91A6193/project-loop>

Live Deployment: <https://project-loop-xi.vercel.app>

1. Project Team & Roles

This capstone project was collaboratively designed, architected, and engineered as a two-member team:

Team Member	Designation	Core Engineering Responsibilities
M. Ranjith Kumar	Full-Stack & AI Lead	Full-stack Next.js 14 architecture, multi-tenant Prisma relational modeling, server-side RBAC security filters, Google Gemini AI grounded RAG engine, API route handlers, and CI/CD Vercel cloud deployment.
M. Renuka Bindu	Frontend & UI/UX Co-Lead	Enterprise dark-mode UI/UX styling, Recharts interactive data visualizations, two-tier responsive navigation, bulk CSV data ingestion pipelines, client-side triage workflow, and system quality assurance.

2. Executive Summary & Problem Statement

Modern digital companies collect feedback from fragmented channels: support tickets, App Store reviews, Google Play ratings, NPS surveys, sales calls, and community forums. Individually, these are isolated comments; collectively, they indicate urgent product fixes and strategic roadmap priorities.

Project LOOP solves this problem by providing a unified multi-tenant intelligence workspace:

- Multi-Channel Ingestion:** Single feedback forms, high-throughput bulk CSV uploads, and simulated real-time channel webhooks (Zendesk, App Store, G2).

- **AI Auto-Classification:** Every incoming customer comment is analyzed in real time for sentiment, confidence scores, urgency level (L1–L5), and category tags.
- **Spike Detection:** Compares week-over-week velocity to alert product teams before minor bugs become major churn events.
- **Ask LOOP (Grounded RAG):** Natural language Q&A engine that answers product questions by citing verbatim customer comments with zero hallucinations.
- **VoC Executive Briefs:** One-click generator producing leadership-ready Voice-of-Customer digests with exportable PDF formatting.

3. System Architecture



4. Technology Stack & Design Rationale

Layer	Technology	Rationale & Benefit
Framework	Next.js 14 (App Router) + TypeScript	Serverless architecture, full-stack route handlers, strict type safety.
Styling	Tailwind CSS + Lucide Icons	High-contrast corporate dark mode theme, responsive 2-tier navigation.
Database & ORM	Prisma ORM + SQLite / PostgreSQL	Declarative relational schema, automatic migrations, strict tenant isolation.

Layer	Technology	Rationale & Benefit
AI Engine	Google Gemini API + Heuristic Fallback	Structured JSON classification, grounded Q&A, and resilient offline capability.
Visualizations	Recharts	Volume gradient area charts, sentiment donut charts, and category bars with 1-click drilldown.
Validation & Parsing	Zod + PapaParse	Runtime API payload validation and high-throughput CSV data ingestion.

5. Key Features & Implementation Highlights

C1 & C2: Multi-Tenancy and Role-Based Access Control (RBAC)

Project LOOP enforces strict workspace data segregation. Every single database query strictly filters by tenantId. The platform features three distinct pre-configured enterprise roles:

- **ADMIN:** Full workspace governance, team member invites, API key management, and VoC report generation.
- **ANALYST:** Feedback triage, CSV imports, manual AI re-classification, and status updates (NEW → REVIEWED → ACTIONED).
- **VIEWER:** Read-only access to analytics and themes. Destructive or write actions trigger HTTP 403 Forbidden toast alerts.

C3 & C4: Feedback Ingestion & Dynamic Triage Workflow

Feedback can be ingested through single-entry forms, simulated webhooks, or CSV spreadsheets. The Inbox supports full-text search, status transitions, and multi-dimensional filtering across channel, sentiment, category, and date ranges with a dedicated one-click filter reset control.

AI1 & AI2: AI Auto-Classification & Trend Spike Detection

Customer submissions are automatically parsed into structured JSON containing sentiment scores, urgency ratings (1 to 5), and actionable business recommendations. Themes with greater than 40% week-over-week velocity growth are automatically flagged in a prominent red Spike Alert Banner.

AI3 & AI4: Ask LOOP (Grounded RAG) & VoC Executive Digests

The Ask LOOP assistant answers natural language questions (e.g., "What are users saying about mobile stability?") by fetching the top relevant customer records and generating a synthesis grounded strictly in verbatim quotes. Leadership teams can generate executive summaries and export them directly to PDF.

6. Demo Verification Credentials

Role	Demo Email	Password	Permissions & Scope
ADMIN	admin@acme.com	password123	Full governance, member invites, VoC reports, triage
ANALYST	analyst@acme.com	password123	Data ingestion, CSV upload, status updates, AI re-classification
VIEWER	viewer@acme.com	password123	Read-only inspection (write operations blocked with HTTP 403)

7. Conclusion

Project LOOP successfully demonstrates a corporate-grade, production-ready full-stack application built for high-throughput customer feedback intelligence. By combining modern React architectures, relational database safety, and intelligent retrieval-grounded AI, the system enables product teams to make confident, data-backed decisions.